

100 Days of Linux - Day 040

Title: Module 4 Practical Workflow & Review - Text Processing

Today's Goal

Complete a realistic text-processing workflow using every command learned in Days 31–39. No new commands are introduced today.

Why This Matters

Cleaning, transforming, and analyzing text files is a daily responsibility for developers, data engineers, and Linux administrators.

Commands Covered

sort, uniq, wc, cut, paste, tr, sed, awk

Workflow Scenario

You received raw employee and sales data. Your task is to clean, validate, merge, and summarize the data before it is loaded into a database.

Practical Assignment (5 Tasks)

1. Create `employees.csv` with the columns ID, Name, Department, Salary and at least 10 records. Use `wc -l` to count the total lines.
2. Extract the Name and Salary columns using `cut`. Then use `awk` to display only employees whose salary is greater than 60000.
3. Create `departments.txt` with duplicate department names. Use `sort` and `uniq -c` to count how many times each department appears.
4. Create `first_names.txt` and `last_names.txt`. Merge them into `full_names.csv` using `paste` with a comma delimiter. Convert the merged output to uppercase using `tr` and save it as `full_names_upper.csv`.
5. Create `application.log` with INFO, WARNING, and ERROR messages. Preview replacing ERROR with CRITICAL using `sed`, apply the change with `sed -i`, verify it with `grep`, and display your current working directory.

Hints

Complete one task at a time. Verify every result with `cat`, `head`, `tail`, `grep`, or `wc` before moving on.

Mini Challenge

Create a `retail_project` containing `employees.csv`, `orders.csv`, `products.csv`, and `application.log`. Use at least six commands from this module to inspect, clean, merge, and summarize the data.

Common Mistakes

Editing files with `sed -i` before previewing the output, or forgetting

to specify the correct delimiter when using cut or awk.

Did You Know?

Many shell scripts combine sort, awk, grep, sed, and wc to automate large-scale log and data processing.

Pro Tip

Use pipes (|) to chain commands together and avoid creating unnecessary temporary files.

Answer Key (Instructor Only)

Q1: `wc -l employees.csv`

Q2: `cut -d ',' -f2,4 employees.csv && awk -F',' '$4>60000 {print $2,$4}' employees.csv`

Q3: `sort departments.txt | uniq -c`

Q4: `paste -d ',' first_names.txt last_names.txt | tr 'a-z' 'A-Z' > full_names_upper.csv`

Q5: `sed 's/ERROR/CRITICAL/g' application.log && sed -i 's/ERROR/CRITICAL/g' application.log && grep 'CRITICAL' application.log && pwd`

Homework

Repeat today's workflow using different sample data until you can complete every task without looking at the answer key.

Next Day Preview

Tomorrow you'll begin the Compression & Archiving module by learning the tar command to package files and directories into archives.